Fiscal Year:	FY 2011	**Task Last Updated:**	FY 01/24/2011
PI Name:	Bouxsein, Mary Ph.D.		
Project Title:	Maintenance of Skeletal Integrity in Spaceflight: Interaction among Partial Weight Bearing, Microgravity and Novel Therapeutics		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	04/01/2010	**End Date:**	03/31/2012
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Despite intensive countermeasures, muscle and bone loss remain among the top medical challenges to extended space missions. Whereas much is known regarding the musculoskeletal responses to microgravity, little is known about the physiological effects of prolonged exposure to partial gravity environments, and whether musculoskeletal deterioration may be mitigated by exposure to partial gravity. To address this knowledge gap, we have developed a novel ground-based model that allows chronic exposure to reduced quadrupedal weightbearing in mice. Using this model will determine whether partial weight bearing attenuates bone and muscle atrophy relative to full hindlimb unloading, and will test the ability of novel spaceflight-compatible therapeutics, one anti-resorptive (antibody to RANKL) and one anabolic (antibody to sclerostin), to inhibit skeletal deterioration across the hypogravity spectrum. Outcome assessments will include in vivo bone density and body composition measurements, along with ex vivo measures of muscle weight, bone microarchitecture, and strength. Serum markers of bone turnover, histomorphometry and gene expression patterns will allow us to delineate the mechanisms underlying the observed skeletal changes. In sum, our proposed studies use both a standardized microgravity analog along with a novel partial gravity model to directly address a key element of the Bioastronautics Roadmap--Accelerated Bone Loss and Fracture Risk. Our ultimate objective is to mitigate the risk for space-induced bone loss and bone fracture associated with missions to low earth orbit, the moon, and eventually Mars. Successful completion of these studies will provide strong rationale for testing these pharmacologic countermeasures in human bed-rest studies and then leading to a trial in astronauts traveling to the International Space Station (ISS). Our findings will also provide insight into the prevention and treatment of skeletal fragility due to immobilization, stroke, cerebral palsy, muscular dystrophy, and spinal cord injury.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The current research has profound impact on Earth-based diseases and disorders resulting from insufficient skeletal loading, such as muscular dystrophy, cerebral palsy, stroke and other neurodegenerative disorders. Our goal is to characterize the amount of skeletal loading necessary to maintain bone and muscle mass, and to determine the efficacy of novel therapeutics in preventing bone and muscle loss. If successful, this would provide novel insight for the rehabilitation of individuals exposed to musculoskeletal unloading or disuse.
Task Progress:	In the first 9 months, we 1) determined the musculoskeletal effects of partial weight bearing in untreated animals; and 2) tested the ability of the anti-sclerostin antibody (Scl-AbII) to prevent bone loss. We studied female wild-type, adult mice (C57Bl/6J, 12 wks of age) subjected to partial weight bearing (PWB20 and PWB40, Lunar and Mars loading analogs) and hindlimb unloading (HLU) for 21 days. Mice were injected with either Scl-AbII at 25 mg/kg or vehicle twice weekly. In vehicle treated animals we observed declines in bone and muscle that were largely dose-dependent for the partial weight bearing animals, including declines in bone and muscle even at 70% weight bearing. Scl-AbII completely inhibited the trabecular and cortical bone deterioration due to disuse, with bone properties in HLU and partial weight bearing mice treated with Scl-AbII being at or above values of fully loaded, vehicle treated mice. As expected, there was no effect of Scl-AbII on muscle loss. For some bone outcomes, the efficacy of Scl-AbII was greater in normally loaded animals than in those exposed to hindlimb unloading or partial weightbearing. Altogether, these early results indicate: 1) partial weight bearing at loading levels up to 70% of body weight is insufficient to prevent bone and muscle loss; and 2) indicate the potent anabolic activity of Scl-AbII, and its efficacy in abrogating disuse-induced bone deterioration. Work still to be completed for these experiments includes histomophometric analyses and serum bone turnover markers, and biomechanical testing. Additional work to be completed in year 2 includes experiments testing the efficacy of a potent anti-resorptive agent (Rank Ligand antibody) if specific genetically engineered mice (hu-RankL) are available. Alternatively, if we cannot gain access to these mice, we propose to test the efficacy of anti-myostatin antibody alone and in combination with anti-sclerostin antibody to prevent musculoskeletal deterioration due to microgravity and partial gravity analog loading.
Bibliography Type:	Description: (Last Updated: 04/02/2026)
Articles in Peer-reviewed Journals	Wagner EB, Granzella NP, Saito H, Newman DJ, Young LR, Bouxsein ML. "Partial weight suspension: A novel murine model for investigating adaptation to reduced musculoskeletal loading." J Appl Physiol. 2010 Aug;109(2):350-7. Epub 2010 Jun 3. PMID: 20522735 , Aug-2010